

SIMATS ENGINEERING

CO5 - ASSESSMENT TOOL 2

Mock Interview

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO5: Students will be able to evaluate advanced machine learning concepts such as reinforcement learning, Monte Carlo methods, sampling techniques, policy iteration, value iteration, and temporal difference learning. They will understand computational learning theory concepts including VC dimension, sample complexity, and mistake-bound analysis. Students will be capable of selecting and optimizing advanced learning strategies for intelligent systems and complex applications. (BL-6)

Assessment Tool Weightage: 20%

QUESTIONS:

Q1. Autoencoder Design

- What is an autoencoder and what are its main components in a neural network architecture?
- How would you design an autoencoder for reducing the dimensionality of a high-dimensional dataset?
- What factors would you consider when deciding the size of the encoder and decoder layers?
- Why is the bottleneck layer important in an autoencoder design?
- How would you choose an appropriate activation function for the encoder and decoder of an autoencoder?

Q2. Stochastic Encoders and Decoders

- What is a stochastic encoder, and how does it differ from a deterministic encoder?
- Why is randomness introduced into the encoding process of a stochastic autoencoder?
- How does a stochastic decoder reconstruct data from a probabilistic latent representation?
- What are the advantages of using stochastic encoders and decoders in generative models?
- How does a stochastic encoder represent uncertainty in the input data?

Q3. Deep Belief Network vs Deep Boltzmann Machine

- What is the main architectural difference between a Deep Belief Network (DBN) and a Deep Boltzmann Machine (DBM)?
- How are connections between hidden layers organized differently in a DBN and a DBM?
- Why is a DBN considered a hybrid model containing both directed and undirected connections?

- How does a DBM differ from a DBN in terms of the direction of connections between layers?
- How does the training procedure of a DBN differ from that of a DBM?

Q4. Generative Adversarial Networks

- What is a Generative Adversarial Network, and what are its two main components?
- How does the generator in a GAN create new data samples? • What is the primary role of the discriminator in a GAN?
- How do the generator and discriminator compete with each other during GAN training?
- Why is balancing the generator and discriminator important for stable GAN training?

Q5. Integrated Deep Learning Solution

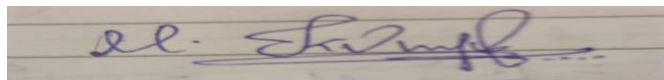
- What do you understand by an integrated deep learning solution, and what are its major components?
- How would you combine different deep learning models to solve a complex real-world problem?
- How would you decide whether to use an autoencoder, DBN, DBM, or GAN in an integrated solution?
- What factors should be considered when designing an end-to-end deep learning pipeline?
- How can feature extraction and representation learning be integrated into a deep learning solution?

Code of Conduct:

I KUMARAN M (192525022) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 Exemplary	Level 3 Proficient	Level 2 Developing	Level 1 Beginning
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts

Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism

ANSWERS

Q1. Autoencoder Design

1. **Autoencoder:** An autoencoder is a neural network that learns to encode input data into a compact representation and then reconstruct the original data.
2. **Dimensionality Reduction:** Use an encoder with progressively smaller layers and a bottleneck layer with fewer neurons than the input features.
3. **Layer Size:** Layer sizes depend on input dimensions, complexity of data, desired compression, and reconstruction accuracy.
4. **Bottleneck Layer:** The bottleneck forces the network to learn the most important features in a low-dimensional representation.
5. **Activation Function:** Choose activation based on the data, such as ReLU for hidden layers and sigmoid/linear for suitable output data ranges.

Q2. Stochastic Encoders and Decoders

1. **Stochastic Encoder:** A stochastic encoder produces a probability distribution or random latent representation instead of one fixed representation.
2. **Purpose of Randomness:** Randomness helps the model represent uncertainty and generate diverse outputs.
3. **Stochastic Decoder:** It samples from the probabilistic latent representation and uses it to reconstruct the input data.
4. **Advantages:** They provide uncertainty modeling, diverse generation, and better representation of complex data distributions.
5. **Uncertainty:** It represents uncertainty by encoding an input as a distribution with parameters such as mean and variance.

Q3. DBN vs DBM

1. **Main Difference:** A DBN has a combination of directed and undirected connections, while a DBM uses only undirected connections.
2. **Hidden Connections:** DBN hidden layers are connected mainly in one direction, whereas DBM layers have bidirectional undirected connections.
3. **Hybrid Model:** DBN is hybrid because its top layers form an undirected RBM while lower layers use directed connections.
4. **DBM Connections:** A DBM has symmetric, undirected connections between adjacent hidden layers.
5. **Training:** DBN is usually trained layer-by-layer using greedy pretraining, while DBM requires joint training of multiple layers.

Q4. Generative Adversarial Networks

1. **GAN:** A GAN is a generative model consisting of a **generator** and a **discriminator**.
2. **Generator:** The generator converts random noise into new synthetic samples similar to real data.
3. **Discriminator:** The discriminator determines whether a sample is real or generated.
4. **Competition:** The generator tries to fool the discriminator while the discriminator tries to correctly identify fake samples.
5. **Balance:** Balance is important so neither network becomes too powerful, allowing stable learning and realistic generation.

Q5. Integrated Deep Learning Solution

1. **Integrated Solution:** It combines multiple deep learning models or techniques to solve different parts of a complex problem.
2. **Combining Models:** Different models can be connected sequentially or in parallel, with each performing a specific task.
3. **Model Selection:** Use an autoencoder for compression, DBN/DBM for feature learning, and GAN for data generation.
4. **Pipeline Factors:** Consider data quality, preprocessing, model compatibility, computational resources, accuracy, and deployment requirements.
5. **Feature Integration:** Feature extraction and representation learning can be integrated by using encoder layers to learn useful features automatically.